

For my capstone, I want to understand mobility friction in New York City spatially. There exists an enormous amount of data documenting these hazards; for example, 22 million 311 complaints since 2020 documenting things like potholes, missing signs, or broken streetlights; 2.2 million vehicle crashes; and 1.5 million double-parking violations (and maybe even locations of Flock cameras?). This project is inspired by NYC Vision Zero and its tools for visualizing where crashes happen across the city. The question, however, is fundamentally different: instead of asking how dangerous a street is, I am asking how difficult it is to move through it. The goal is to visualize these frictions spatially, with an emphasis on the presentation/dashboard. The benefit of this analysis will be an equity lens—what neighborhoods, demographic groups, and socioeconomic communities experience the most hazards? Where is mobility most difficult relative to the use of the street?

Sources:

[311 Service Requests from 2020 to Present](#)

[Motor Vehicle Collisions - Crashes](#)

[Parking Violations Issued - Fiscal Year 2027](#)

[Traffic Volume Counts \(Historical\)](#)